Anim. Behav., 1971, **19**, 174–191

EFFECTS OF 6 DAYS SEPARATION FROM MOTHER ON 18- TO 32-WEEK-OLD RHESUS MONKEYS

BY YVETTE SPENCER-BOOTH & R. A. HINDE

Sub-Department of Animal Behaviour and Medical Research Council Unit on the Development and Integration of Behaviour, Madingley, Cambridge

Abstract. Nineteen group-living infants were subjected to separation when 18, 21, 25 to 26 or 30 to 32 weeks old. The 21 and 25 to 26 week groups were separated again at 30 to 31 weeks. Effects on various activity measures and the mother–infant relationship were assessed quantitatively. During separation activity and play decreased. When the mother returned the infants were more clinging than before separation. Some of the effects of separation were still apparent 4 weeks later. The sex of the infants, contact with group companions during separation and age at separation did not account for variability in the response to separation. Infants separated for the second time responded similarly to infants separated for the first time at the same age.

It is now well established that human children may be profoundly disturbed by a period of separation from the mother (e.g. Bowlby 1952; Lebovici 1962; Ainsworth 1962). However the dynamics of the disturbance, and the marked individual differences in the severity of the effects, are far from being understood (e.g. Ainsworth 1962). In part this is due to the ethical impossibility of relevant experimentation: the possibility therefore arises that suggestive leads could be obtained from sub-human species. A number of studies have in fact demonstrated that relatively brief separation from the mother can produce at least short-term effects in monkeys (Jensen & Tolman 1962; Seay, Hansen & Harlow 1962; Seay & Harlow 1965; Spencer-Booth & Hinde 1967; Kaufman & Rosenblum 1967a, b, 1969; Rosenblum & Kaufman 1968). The data presented here were obtained in an attempt to specify the symptoms quantitatively, to determine the nature and cause of individual differences in the response to a separation experience, and to assess its long-term effects.

The experiments involved four groups of rhesus monkeys whose mothers were removed for 6 days when they were about 18, 21, 25 and 30 weeks old. Most infants in the 21- and 25-week-old groups were subjected to a second separation experience at about 30 weeks. In this way we attempted to assess the extent to which the responses were affected by age at separation and by repeated separation experiences. Data from four of the five animals in the 30-week group were reported previously (Spencer-Booth & Hinde 1967). This paper is concerned with the nature of the immediate response to separation, and its effect in the ensuing weeks. Other aspects of this work will be discussed in later papers.

Methods

Material

The infants were reared in groups each containing an adult male, two to four females and their young, kept in outdoor cages 540 × 240 × 240 cm) connecting with indoor rooms (180 × 135 × 225 cm: see Hinde & Rowell 1962). The experimental groups contained nineteen infants, divided between groups as follows: 18 weeks, one male and two female infants; 21 weeks, two male and four female infants, five of which were separated again at 30 weeks (one female infant died during the first experiment); 25 to 26 weeks, four male and one female infants, all separated again at 30 to 31 weeks; 30 to 32 weeks, two male and three female infants.

Methods

The separation experiments involved removing the mother from the group and placing her in an indoor room about 54 m from and out of sight of the group for 6 days, after which she was returned. One 21-week infant and one 26-week infant collapsed after 2 and 5 days separation respectively, whereupon the mothers were returned prematurely: both infants survived. Data were collected as follows:

Pre-experimental period. Routine watches were made on mother–infant interaction from soon after birth up to the experimental period. Watches were made between 09.00 and 13.00 hours: procedural details are given in Hinde,

Rowell & Spencer-Booth (1964). Each infant was watched for at least 6 hr every other week (in many cases every week) except for one infant in the 26-week group which was not introduced to the colony until 9 weeks old.

Experimental period. *Pre-separation period.* Each infant was watched on 3 of the 4 days (termed days 1 to 3) before the mother was removed, starting at 10.00 hours (two of the 30- to 32-week infants were watched only on 2 days). Routine data were collected as previously, and the infant's activity while off its mother was recorded (see Spencer-Booth & Hinde 1967). The watch was continued until the infants had been off the mother and all other animals for a total of 2 hr, or until 14.30 hours, whichever was the earlier.

Separation period. 1t 10.00 hours on day 4 of the experiment the mother was removed. The infant was watched as on days 1 to 3 on this day, and on days 5, 7 and 9.

Post-separation period. At 10.00 hours on day 10 the mother was returned (see above for exceptions). Data collection began immediately and as on days 1 to 3 continued until the infant had been off its mother and all other animals for 2 hr or until 14.30 hours, whichever was earlier. Similar watches were made on days 11, 12, 14, 16, 23, 30 and 37. For the two infants whose mothers were returned early, the day of her return was counted as day 10 for the purpose of analysing post-separation data.

The 21-week and 25 to 26-week groups were given a second separation experience at 30 to 31 weeks. For the 25 to 26 week group this meant that day 37 of the first separation was also one of the three pre-separation days for the second.

Post-experimental period. Routine watches were continued as previously for 6 hr every 4 weeks (the 6 hr being all in one week) until the infants were a year old, except for two infants which died (see below). These data will be discussed in a later paper.

Tests. The infants were given certain tests to assess individual behaviour characteristics. All were tested during the initial pre-separation period and on day 5 of the first separation. Thereafter the 21-week group were tested on day 37, again during their second pre-separation period at 30 weeks, and on days 5 and 37 of their second separation experiment. The 25 to 26-week group had further tests on day 37 of the first separation experiment (i.e. during the pre-separation period of their second separation). They were tested again on days 5 and 37 of the second separation experiment.

Six observers were involved in the separation experiments. All observers were extensively trained beforehand, and the categories of behaviour used were so clear-cut that inter-observer differences were negligible and an assessment of inter-observer reliability unnecessary.

Measures

The measures obtained from the routine and activity watches fall into a number of groups.

Measures of the mother–infant relationship. These were obtained from the routine watch data except *percentage following* which came from activity watch sheet (see d).

(a) *The proportion of time the infant spent on or off the mother. Asleep.* Percentage of ½-min in which the infant was on the mother with its eyes closed. *Awake on nipple.* Percentage of ½-min watched in which the infant was thus. *Off nipple.* Percentage ½ min watched in which the infant was on the mother but not holding the nipple. *Time off.* Percentage ½ min watched in which the infant was off the mother and all other animals. In some cases time off mother is referred to, which includes time spent on group companions. *On and off.* Percentage of ½ min in which the infant was off the mother in which it was also on her. This is inversely related to the length of bouts off the mother.

(b) *Measures relating to the responsibility for the infant being on or off.* Every time the infant got on or tried to get on the mother we recorded whether it was through the mother's initiative (M) or the infant's: in the latter case the infant could be accepted (A) or rejected (R). This provides two measures: *R.* The mean number of rejections by the mother per hundred ½ min watched. $R/(A+M+R)$. The proportion of the total number of times the infant got on the mother, or attempted to do so, which were rejected.

(c) *Measures of where the infant was when it was off the mother.* >60 *cm.* The percentage of ½ min in which the infant was off in which it was recorded more than 60 cm from the mother (60 cm is roughly the distance a mother can reach in a hurry, and was assessed in relation to the 120 cm divisions of the wire netting). >60 *cm only.* The percentage of ½ min in which

the infant was off in which it remained more than 60 cm from the mother throughout the ½ min. *Leaning against mother.* The percentage of ½ min off in which the infant was in close physical contact with the mother while off.

(*d*) *Measures of responsibility for maintaining proximity. Percentage Ap.* The percentage of occasions that the distance between mother and infant decreased from more than 60 cm to less than 60 cm (approaches) which were due to movement by the infant. *Percentage L.* The percentage of occasions that the distance between mother and infant increased from less than 60 cm to more than 60 cm (leavings, which were due to movement by the infant. *Percentage Ap.–Percentage L.* The difference between the proportions of approaches and of leavings which were due to movement by the infant. This gives a measure of initiative in maintaining proximity with the mother which is relatively independent of the relative activity of the two animals (Hinde & Atkinson 1970). *Percentage F.* The percentage of ½ min in which the infant was off its mother in which it was recorded as following the mother around the pen.

Measures of the infant's relationship with group companions. *Leaning against.* The percentage of ½ min off in which the infant was in close physical contact with a group companion other than the mother. *Time on group companions.* The percentage of ½ min watched in which infants were recorded as clinging ventro-ventrally to group companions other than the mother.

Vocalizations. Obtained from routine watch sheets. *Whoos.* The mean number of whoo calls (the calls particularly noticeable in separated infants) per hundred ½ min: (a) in which the infant was off, (b) in which the infant was on the mother, (c) in which the infant was on a group companion. Squeak/gecker/screech (S/G/S). This assembly of intergrading noises is given by infants being rejected by their mothers and also by separated infants. They were analysed in terms of the percentage of ½ min watched in which any of them were recorded. *Tantrums.* These consist of loud and intense screeching, often accompanied by convulsions, and are quite frequent among infants being rejected by their mothers, but they may also occur in her absence. They were analysed in terms of the percentage of ½ min watched in which they were recorded.

Measures of amount of activity. These were obtained from the activity watch sheets, and are thus expressed in relation to the time spent off all animals. *Locomotor activity.* The number of 'boxes' entered by the infant per ½ min in which it was off (the cage was considered as divided into sixteen sections). *Activity when active* ($\frac{2\text{–}4}{>1}$ per cent). The number of ½ min in which the infant entered two to four boxes, as a percentage of the number in which it entered more than one. The higher the percentage, the less active the infant when active.

Measures of the type of activity. *Percentage S.* The number of ½ min in which the infant was recorded sitting still and not engaged in any obvious activity, as a percentage of the number in which it was off. *Percentage E.* The percentage of ½ min of the whole watch (i.e. including the time it was on the mother or another animal) in which it was recorded as eating either from supply or pouches. *Percentage SF.* The number of ½ min in which the infant was engaged in manipulative activity, such as sitting fiddling with pieces of wire or chain, as a percentage of the number it was off. *Percentage R&T.* The number of ½ min in which the infant was engaged in rough-and-tumble social play, as a percentage of the number it was off. *Percentage A/W.* The number of ½ min in which the infant was engaged in approach/withdrawal social play, as a percentage of the number it was off.

Method of Analysis

For each animal the data were grouped as follows: the three pre-separation days; day 4 (mother's removal); days 4 to 9 (separation mean, including day 4); day 10 (mother's return); days 11 to 16 and days 23 to 37. Where watches were lumped, individual means were used in the analysis. There are of course no data for mother–infant interaction during separation, and some infants were off their mothers so little on day 10 that measures relative to time off mother were unreliable.

Differences between age groups and sexes turned out to be slight (see pp. 183–188). In the initial analysis, aimed at establishing the nature of the effects of the separation experience, the data were therefore considered in two categories. First all infants were assessed at their first separation (21-, 25- and 30-week groups), and secondly all second separation animals were assessed. The effects of each separation were

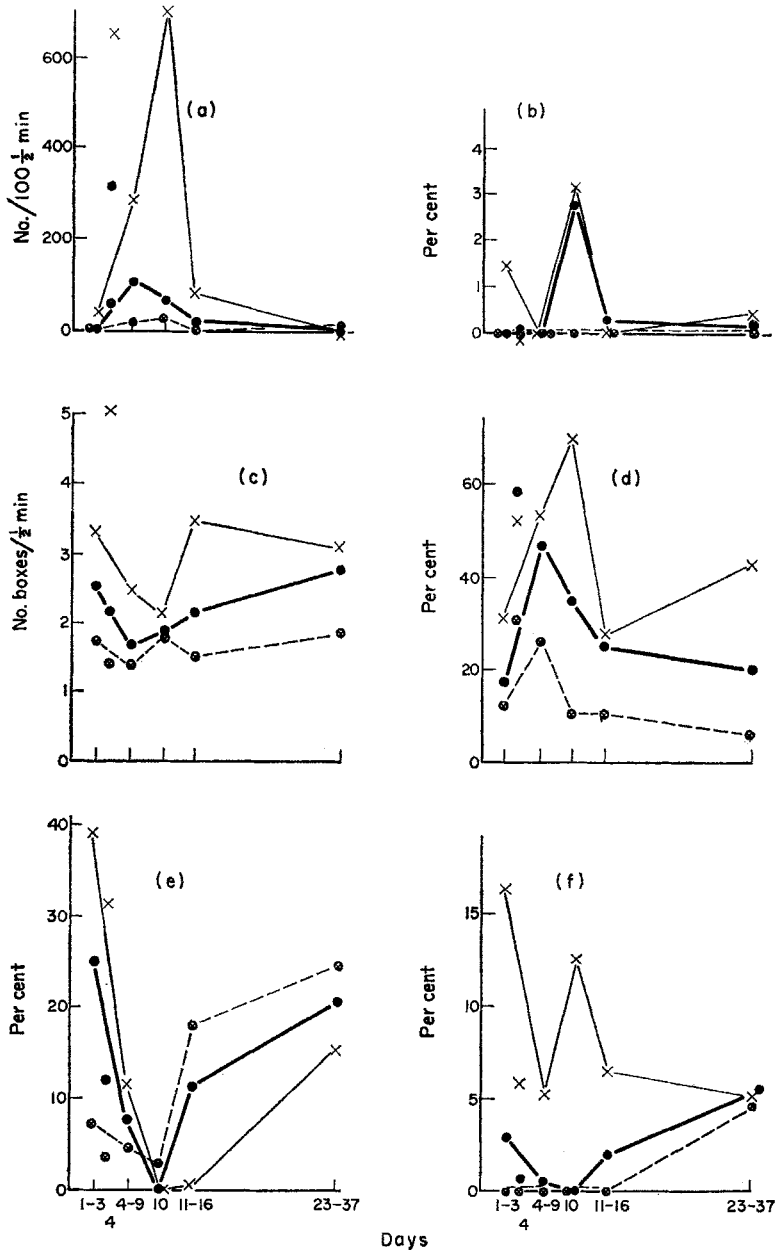


Fig. 1. The effects of separation on infant's behaviour. (a) Number of whoo-calls per 100 ½-min off all animals. (b) Percentage of ½-min off all animals in which tantrummed. (c) Activity: Number of boxes per ½-min. (d) Percentage of ½-min off in which sat. (e) Percentage of ½-min off in which showed manipulative play. (f) Percentage of ½-min off in which showed rough and tumble social play. ●, Median for all first separation infants; ×, Individual score for first separation infant with the highest pre-separation mean; ⊗, Individual score for first separation infant with the lowest pre-separation mean. Days 1–3 = pre-separation mean; day 4 = day of mother's removal; days 4–9 = separation mean; day 10 = day of mother's return; days 11–16 = mean of days 11, 12, 14, 16; days 23–37 = mean of days 23, 30, 37.

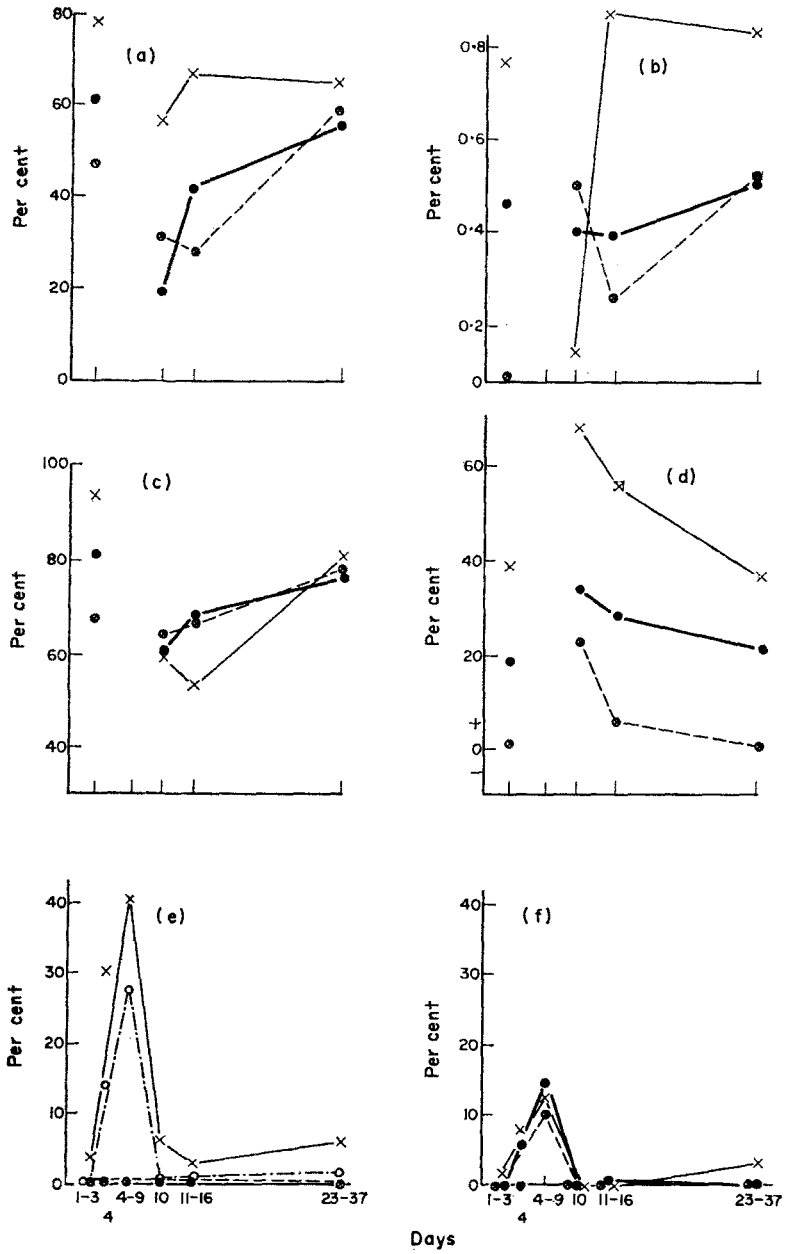


Fig. 2. The effects of separation on the relationship of the infants with their mothers and with group companions. (a) Percentage of ½-min watched in which off all animals (time off). (b) Ratio of number of times infants were rejected to total number of times infants attempted to gain nipple or were picked up by their mothers (R/A+M+R). (c) Percentage of ½-min in which infants were off all animals in which they were >60 cm from their mothers (>60 cm). (d) Difference between percentage of approaches (from more than 60 cm to less than 60 cm) which were due to movement by infants, and percentage of leavings (from less than 60 cm to more than 60 cm) which were due to infants (%Ap.-%L). (e) Percentage of ½-min watched in which infants were recorded in ventro-ventral contact with group companions (on group companions). (f) Percentage of ½-min watched in which infants were recorded leaning against group companions. Conventions as in Fig. 1, except in (e) where the median was always zero, and has been omitted, and ○—— · ——○ is the individual score for an infant with an intermediate pre-separation mean score.

assessed in the first place by comparison of individual mean pre-separation scores with those in each separation and post-separation period (Tables I and II), the overall significance being assessed by Wilcoxon matched pairs signed ranks tests (two-tailed). This assumes the justifiability of ranking differences between pre-separation and subsequent data, without regard to initial level. Sign tests were therefore also used, and Tables I, II and III show where a probability level of $P<0{\cdot}05$ was indicated by one test and not the other. Medians for the more important measures for first separation infants are shown in Figs 1 and 2, which also show the individual scores for the two infants at the extremes of the range of pre-separation scores.

Infant Mortality

The three 18-week infants died in the period following the mothers' return. At post-mortem one (died 19 days after mother's return) had a congenital patent foramen ovale; one (16 days) had an abdominal retroperitoneal mass (probably a carcinoma) and in the third (5 days) the cause of death could not be ascertained. Data from this group are not considered further. One infant from the 21-week group died a week after the mother's return: post-mortem examination revealed no evidence of infectious or parasitic disease. This was not one of the infants whose mothers were returned early. We are unable to say how far the deprivation experience contributed to the deaths of these infants.

Two other infants died before they were a year old, one from the 21-week group at 50 weeks and one from the 26-week group at 47 weeks. Post-mortem examination of the former revealed a generalized stomatitis with ulceration of the cheek and lip due to *Monilia*, and of the latter acute pasteurella pneumonia. A further animal from the 25-week group died at 110 weeks: post-mortem examination showed extensive pneumonia; no organisms could be cultured.

Changes During and After Separation

Vocalizations (Table I; Fig. 1)

Tantrums and squeaks/geckers/screeches (S/G/S) on the mother were rare, and have not been considered here.

On the day the mother was removed (day 4) there was a large increase in whoo calls and a decrease in tantrums: S/G/S increased in first separations and decreased in second separations. Similar changes were evident when the separation period was considered as a whole.

On the day the mother was returned (day 10) all three types of vocalization were more frequent than before separation. Tantrums ($P<0{\cdot}01$ and S/G/S (1st separation, $P<0{\cdot}02$; 2nd separation $P<0{\cdot}01$) were also higher than the separation mean. On days 11 to 16 levels were still higher than before separation (significantly so for first separations only), but on days 23 to 37 the difference was still significant only for the whoos of first separation infants.

Measures of Activity (Table I; Fig. 1)

As described previously, infants whose mothers have recently been removed adopt a hunched posture and behave as though depressed. Over the separation period as a whole the infants showed less locomotor activity, were less active when active (i.e. $\frac{2\text{–}4}{>1}$ percentage increased), and were recorded sitting more often. They also played less and ate more. Although the change in time spent sitting was evident on the day the mother was removed (day 4), those in locomotor activity and activity when active did not appear until later in the separation period.

All these changes in activity measures were still present on the day the mother was returned, but were ameliorated on days 11 to 16 and had disappeared in most infants by days 23 to 37.

Measures of Relationship with Group Companions (Table I; Fig. 2)

The age and sex categories of social companions available differed between infants, but all categories have been known to cuddle infants at least sometimes (Spencer-Booth 1968). Here we shall consider changes in time spent leaning against or on group companions. The categories of group companions which accepted the infants, and the effects of having been accepted, will be discussed later.

Since only some of the infants leant against and went on group companions, the medians were often zero. In Table I the medians refer to all infants in the groups, but the proportion showing a change refers only to the animals scoring in at least one of the periods being compared. No data for leaning against group companions were available for four of the 30-week infants.

On the day the mother was removed (day 4), and even more over the separation period as a whole, a considerable number of infants scored

more highly for leaning against and time on social companions. This tendency for increased interaction with group companions was still discernible on day 10, but diminished thereafter.

Measures of Mother–Infant Relationship (Table II; Fig. 2)

Proportion of time on or off mother. Immediately after the mother's return the infants spent more time on their mothers. Time asleep increased, and times off mother and off all animals decreased, in both first and second separation categories: time awake on nipple increased in the first but not the second.

On days 11 to 16 considerable recovery towards pre-separation levels had occurred, though in 1st separation tests time asleep was still greater and times off mother and off all adults still lower. In second separations most animals spent less time awake on nipple, the opposite trend to that seen in first separations. By days 23 to 37 none of these measures showed significant differences from pre-separation levels.

Measures relating to responsibility for time on or off. On day 10 the absolute frequency of rejections was lower than before separation in first separation animals, but the change in relative frequency was not significant. There were no significant differences from pre-separation levels for day 10 of second separations, nor for either category on days 11 to 16 or 23 to 37. However with these measures lumping of age groups concealed significant differences between them: we shall see later that the absolute and relative frequencies of rejections were decreased in the 21-week group on day 10 and the absolute frequency on days 11 to 16, but older age groups showed a smaller or reversed change.

Proximity when off mother. At both first and second separations the percentage of ½ min for which the infant was off all animals in which it was recorded more than 60 cm from its mother was smaller on day 10, days 11 to 16 and days 23 to 37 than before separation, though the difference for second separations on days 23 to 37 was not significant. The changes in the percentage of complete ½ min spent more than 60 cm from the other were in a similar direction, but less marked: after separation there was also a tendency for infants to spend a higher proportion of the time off in close proximity to her, but in no case was the difference significant.

Responsibility for proximity. On day 10 the infants were relatively less active than before separation, the percentage of both approaches and leavings due to the infants being less than before separation for both first and second separations. This change was subsequently reversed: on days 11 to 16 the reduction in the percentage of approaches was no longer significant, and on days 23 to 37 it had become a significant increase for the first separation infants, the percentage of leavings showing no overall change from pre-separation levels.

The infant's role in the maintenance of proximity (%Ap.–%L) increased in most infants on day 10, though the difference was not significant: this was partly due to the number of infants which were so seldom more than 60 cm from their mothers on this day that %Ap.–%L could not be assessed. The median %Ap.–%L was slightly less on days 11 to 16 than on day 10, but the difference from pre-separation levels was now significant for both first and second separations. By days 23 to 37 it had returned to near the pre-separation levels.

Taken together these results indicate that after the mother's return the infants showed less locomotor activity in either approaching or leaving her: this was presumably an aspect of the reduction in general motor activity (see above). However they played a greater role (i.e. %Ap.–%L more) in maintaining proximity to her: this is in harmony with the trend towards an increase in the frequency with which the infants followed their mothers throughout the post-separation period, though the latter never reached significance.

Changes Within the Separation Period (Table III)

We have seen that the reduction in locomotor activity, characteristic of the separation period as a whole, was not evident on the day the mother was removed. Here we consider in more detail the changes in the infants' behaviour during the separation period. Inspection of the data showed that changes were progressive throughout the separation period: comparisons have therefore been made only between the data for day 4 and day 9 (Table III). The two infants returned early had no day 9 scores, and some infants had no score on either day for some measures. All these have been omitted from the proportions showing a change.

The time spent leaning against group companions tended to increase during the separation period (significantly only for second separations) and time spent on group companions tended to increase for those infants which went

Table I. Vocalizations, Relations with Group Companions and Activity Before, During and After Separation

		Pre separation median	Day 4 median	Day 4 vs pre-sep ±	Day 4 vs pre-sep no.	Day 4 vs pre-sep P	Separation mean median	Separation vs pre-sep. ±	Separation vs pre-sep. no.	Separation vs pre-sep. P	Day 10 median	Day 10 vs pre-sep. ±	Day 10 vs pre-sep. no.	Day 10 vs pre-sep. P	Days 11–16 median	Days 11–16 vs pre-sep. ±	Days 11–16 vs pre-sep. no.	Days 11–16 vs pre-sep. P	Days 23–37 median	Days 23–37 vs pre-sep. ±	Days 23–37 vs pre-sep. no.	Days 23–37 vs pre-sep. P
(1) Vocalizations																						
Whoos/100 ½ min																						
(a) Off all	(i)	2·6	316	+	16/16	0·01	110	+	16/16	0·01	62	+	12/13	0·01	17	+	13/15	0·01	9	+	11/14	0·05
adults	(ii)	0·7	116	+	9/10	0·01	45	+	8/10	(0·05)	47	+	8/9	0·02	6	+	8/10	NS	2	−	5/9	NS
(b) On mother	(i)	0									1	+	9/10	0·01	1	+	9/10	0·02	0	+	4/5	NS
	(ii)	0									0	−	1/2	NS	0	−	1/1	NS	0	+	2/10	NS
Squeaks/geckers/	(i)	1·03	2.93	+	12/16	(0·05)	2·64	+	12/16	NS	14·4	+	12/14	0·01	4·6	+	13/16	0·01	1·94	+	9/15	NS
screeches off	(ii)	1·91	0·42	−	9/10	0·05	0·41	−	9/10	0·05	13·85	+	8/9	0·01	2·53	−	6/10	NS	2·19	−	6/10	NS
Tantrums off	(i)	0	0	−	8/9	0·05	0	−	6/11	NS	2·8	+	12/14	0·01	0·28	+	11/14	NS	0·16	+	6/12	NS
	(ii)	0·4	0	−	8/10	NS	0	−	8/10	(0·02)	6·4	+	8/9	0·01	0·4	+	6/9	NS	0·13	−	6/9	NS
(2) Group companions																						
Time leaning	(i)	0·25	6	+	9/10	0·01	15	+	12/12	0·01	0	+	4/8	NS	0·25	+	5/10	NS	0	+	4/9(1=)	NS
against	(ii)	0·1	2	+	7/8	(0·05)	7	+	10/10	0·01	0	+	4/6	NS	0·75	+	6/8	NS	0·1	−	4/7	NS
Time on	(i)	0	1	+	7/7	0·05	4	+	9/9	0·01	0	+	6/6	0·05	0	+	6/7	NS	0	+	5/5	NS
	(ii)	0	0	+	2/3	NS	0	+	4/4	NS	0	+	4/4	NS	0	+	3/4	NS	0	+	1/2	NS
(3) Locomotor activity																						
No. boxes/½ min	(i)	2·6	2·1	−	11/16	NS	1·7	−	15/16	0·01	1·8	−	12/13	0·01	2·1	−	12/16	(0·01)	2·7	−	8/15	NS
	(ii)	2·4	2·5	+	6/10	NS	2·0	−	8/10	(0·02)	1·8	−	9/9	0·01	2·0	−	8/10	NS	2·4	+	7/10	NS
2−4 / >1 %	(i)	77	79	−	9/16	NS	86	+	14/16	0·01	88	+	11/13	(NS)	85	+	13/16	0·01	75	−	9/15	NS
	(ii)	81	79	−	7/10	NS	87	+	6/10	NS	86	+	8/9	0·05	84	+	7/10	NS	78	−	9/10	(NS)
% Sitting	(i)	18	58	+	16/16	0·01	47	+	16/16	0·01	35	+	7/13	NS	25	+	8/16	NS	20	+	9/15	NS
	(ii)	23	51	+	10/10	0·01	34	+	9/10	0·01	39	+	8/9	(0·05)	29	+	6/10	NS	22	+	5/10	NS
(4) Type of activity																						
% eating	(i)	30	27	−	8/16	NS	40	+	12/16	(0·02)	5	−	14/16	0·01	25	−	10/16	NS	31	−	8/15	NS
	(ii)	25	45	+	8/10	NS	51	+	10/10	0·01	16	−	10/10	0·01	36	+	8/10	(0·01)	34	+	9/10	(NS)
% Sitting	(i)	25	12	−	14/16	0·01	8	−	16/16	0·01	0	−	13/13	0·01	12	−	14/16	0·01	21	−	9/15	NS
fiddling	(ii)	20	9	−	9/10	0·01	7	−	9/10	0·01	2	−	9/9	0·01	15	−	6/10	NS	19	−	7/10	NS
% R&T play	(i)	2·9	0·6	−	11/15	NS	0·5	−	14/15	0·01	0	−	10/12	0·01	2·1	−	9/15	NS	5·4	+	8/15	NS
	(ii)	6·1	1·1	−	10/10	0·01	0·6	−	10/10	0·01	0	−	9/9	0·01	3·1	−	9/10	0·05	4·5	−	7/10	NS
% A/W play	(i)	0·9	0	−	10/13	NS	0	−	12/15	0·01	0	−	9/13	0·01	0·6	−	9/14	NS	3·3	+	11/14	(0·02)
	(ii)	1·9	0	−	9/10	0·02	0	−	10/10	0·01	0	−	8/9	0·01	0·7	−	8/10	(0·05)	1·0	−	6/10	NS

Medians for (i) first separation infants from groups separated at 21 to 32 weeks and (ii) second separation infants. ± indicates direction of change from pre-separation level shown by number of individuals indicated in next column. (1=) indicates no change in one infant. *P* value from Wilcoxon two-tailed test: this is enclosed in parentheses if Wilcoxon gives the significance level indicated when the sign test did not, or vice versa. Only the infants showing behaviour in either period under consideration are included in proportions. Medians based on all infants. NS=non-significant.

Table II. Mother–Infant Relationship Before and After Separation

		Pre-separation median	Day 10 median	Day 10 vs pre-sep. ±	Day 10 vs pre-sep. no.	Day 10 vs pre-sep. *P*	Days 11–16 median	Days 11–16 vs pre-sep. ±	Days 11–16 vs pre-sep. no.	Days 11–16 vs pre-sep. *P*	Days 23–37 median	Days 23–37 vs pre-sep. ±	Days 23–37 vs pre-sep. no.	Days 23–37 vs pre-sep. *P*
(1) Proportion of time on or off														
(a) Asleep	(i)	21	46	+	14/15	0·01	31	+	11/16	(0·05)	20	−	8/15	NS
	(ii)	21	43	+	9/10	0·01	21	+	5/10	NS	19	+	6/10	NS
(b) Awake on nipple	(i)	20	27	+	12/15	0·01	21	+	9/16	NS	24	+	10/15	NS
	(ii)	25	28	+	6/10	NS	15	−	9/10	0·01	22	−	7/10	NS
(c) On mother off nipple	(i)	3·7	6·0	+	9/15	NS	3·9	+	10/16	NS	4·7	+	10/15(1=)	NS
	(ii)	3·2	3·0	+	6/10(1=)	NS	2·3	−	8/10	NS	2·7	−	8/10	NS
(d) Time off all animals	(i)	60	18	−	16/16	0·01	41	−	11/16	(0·01)	55	−	10/15	NS
	(ii)	59	26	−	10/10	0·01	64	+	6/10	NS	65	+	7/10	NS
Time off mother	(i)	61	19	−	15/16	0·01	42	−	11/16	(0·01)	55	−	10/15	NS
	(ii)	59	28	−	10/10	0·01	65	+	6/10	NS	65	−	3/10	NS
(e) On and off	(i)	9	20	+	10/14	NS	14	+	11/16	NS	10	+	9/15	NS
	(ii)	10	13	+	6/9	NS	7	−	8/10	NS	7	−	7/10	NS
(2) Responsibility for time off														
(a) R	(i)	2·9	1·1	−	13/15	0·02	2·2	−	11/16	NS	3·1	+	8/15	NS
	(ii)	3·1	2·0	−	7/10	NS	3·3	+	7/10	NS	3·2	−	6/10	NS
(b) R/(A+M+R)	(i)	0·46	0·40	−	10/15	NS	0·39	−	10/16	NS	0·50	+	10/15	NS
	(ii)	0·53	0·51	−	6/10	NS	0·55	+	6/10	NS	0·42	−	6/10	NS
(3) Distance when off														
(a) > 60 cm	(i)	81	60	−	11/14(1=)	0·01	68	−	15/16	0·01	76	−	12/15	0·01
	(ii)	75	55	−	8/9	0·02	68	−	9/10	0·05	70	−	7/10	NS
(b) > 60 cm only	(i)	47	24	−	11/14	(0·05)	36	−	12/16	(0·01)	40	−	12/15	(NS)
	(ii)	41	22	−	8/9	(NS)	38	−	7/10	NS	36	−	6/10	NS
(c) Leaning against mother	(i)	2	5	+	5/9	NS	6	+	9/11	NS	3	+	5/10	NS
	(ii)	5	9	+	7/9	NS	5	+	6/10	NS	5	−	6/10	NS
(4) Responsibility for proximity														
(a) %Ap.	(i)	86	66	−	9/10	0·01	83	−	10/16	NS	92	+	11/14	(0·01)
	(ii)	89	70	−	8/8	0·01	87	−	5/10	NS	93	+	6/10	NS
(b) %L	(i)	69	47	−	11/12	0·01	49	−	13/16	0·01	56	+	7/15	NS
	(ii)	64	30	−	7/8	(0·02)	47	−	9/10	0·01	64	−	6/10	NS
(c) %Ap.–%L	(i)	19	34	+	8/10	NS	28	+	12/16	(0·02)	21	+	8/15	NS
	(ii)	21	40	+	6/8	NS	31	+	9/10	0·01	26	+	7/10	NS
(d) %F	(i)	2·8	3·8	+	8/13	NS	5·06	+	13/16	(NS)	4·16	+	10/15	NS
	(ii)	4·7	5·5	+	6/9	NS	6·11	+	8/10	NS	5·63	+	8/10(1=)	(NS)

Conventions as Table I.

Table III. Relations with Group Companions, Vocalizations and Activity at the Beginning and End of Separation Period

		Day 4 median	Day 9 median	Day 9 vs day 4		
				±	no.	*P*
(1)						
Time leaning against	(i)	6	17	+	5/9	NS
	(ii)	2	8	+	7/9	0·05
Time on	(i)	1	7	+	6/8	0·05
	(ii)	0	0	+	3/4	NS
(2) Vocalizations						
Whoos off all adults	(i)	316	9	−	14/14	0·01
	(ii)	116	2	−	10/10	0·01
s/g/s	(i)	2·93	2·9	−	8/14	NS
	(ii)	0·42	0·2	−	5/10(3=)	NS
(3) Locomotor activity						
No. boxes /½ min	(i)	2·13	1·61	−	12/14	0·01
	(ii)	2·47	1·75	−	9/10	0·01
$\frac{2\text{–}4}{>1}$ %	(i)	79	90	+	13/14	0·01
	(ii)	79	99	+	9/10	0·01
% sitting	(i)	58	43	−	9/14	NS
	(ii)	51	24	−	9/10	0·01
(4) Type of activity						
% eating	(i)	27	51	+	10/14	0·01
	(ii)	45	48	+	6/10	NS
% sitting fiddling	(i)	12	5	−	10/14	0·05
	(ii)	9	9	−	6/10	NS
% R&T play	(i)	0·6	0	−	7/8(1=)	0·05
	(ii)	1·1	0	−	5/8	NS
% A/W play	(i)	0	0	−	3/3	NS
	(ii)	0	0	−	3/3	NS

Conventions as Table I.

on at all (significantly only for first separations). Some of the signs of depression increased as the separation period progressed. Thus locomotor activity and activity when active decreased, in both first and second separations. Few animals played socially during separation, but those which did so played less at the end of the period than at the beginning. However the frequency of whoo-calls decreased as the period progressed, and there were signs of recovery in the tendency to sit less (per cent S, significantly only for second separations), to sit fiddling) less significantly only for first separations) and to eat more (significantly only for first separations) towards the end of the period.

Thus the changes in the infant's behaviour which are initiated at, or shortly after, the mother's removal, with the exception of sitting and whoo-calling, progressed throughout the 6 days and there was thus little indication of recovery during this period of maternal deprivation.

Some Factors Possibly Affecting the Response to Separation

Sex Differences

Although the overall sex ratio was equal, the several age groups did not contain equal numbers of males and females. Male infants tend to be on their mothers more than female infants in the early weeks, and less after about 30 weeks (Hinde & Spencer-Booth 1967, 1970); these separation experiments thus took place in the transition period. Those measures in which the sex difference approached significance (two-tailed) in any one period are shown in Table IV.

Although none of the differences before separation quite reached significance, males tended to be off their mothers less, to receive a higher proportion of rejections, and to play a larger role in maintaining proximity, than females. During the post-separation period the median time off the mother remained less for males, but none of the differences were significant. The sex difference in the relative pro-

Table IV. Median Values of Measures which Showed Differences Approaching Significance (Mann–Whitney U, two-tailed) Between Males and Females in One of the Observation Periods

Measure		Pre-sep	Sep	Day 10	Days 11–16	Days 23–37
Time off	Male	59		17	41	56
	Female	64		28	47	64
	P	0·08		NS	NS	NS
R/(A+M+R)	Male	55		41	41	52
	Female	35		44	39	27
	P	0·08		NS	NS	0·01
>60 cm	Male	81		78	71	74
	Female	82		56	66	76
	P	NS		0·12	NS	NS
%Ap.–%L	Male	22		45	36	33
	Female	13		23	14	13
	P	0·13		0·09	0·04	0·004
%L	Male	69		10	38	57
	Female	70		53	54	80
	P	NS		0·05	NS	0·15
% sitting	Male	27	50	69	28	29
	Female	17	43	13	17	15
	P	NS	0·13	0·05	0·03	0·02
Whoos	Male	3	137	146	12	9
	Female	4	99	11	17	1
	P	NS	NS	0·05	NS	0·15

portion of rejections was reduced in the immediately post-separation period, but was re-established on days 23 to 37. When off their mothers' the males tended to be more often at a distance from them in the immediately post-separation period, but not later. However the difference in the role of the infant in maintaining proximity became progressively more significant through the post-separation period: this appears to have been due primarily to males being responsible for a smaller proportion of leavings. The median number of ½-min in which the infant was recorded sitting was (non-significantly) higher for males than for females before separation: the difference increased on day 10, and remained significant through the separation period. Males also whoo-called more than females on day 10.

In summary, these data, though not conclusive, suggest that males tend to be more upset by such a period of separation from their mothers than females, and that this may be a consequence of differences between the relationship of male and female infants with their mothers.

No sex differences in relations with social companions other than the mother were found.

Role of Group Companions

Nine of the sixteen first separation infants, and four of the ten second separation infants, were recorded on group companions during the mother's absence. In addition, many infants spent a considerable amount of time leaning against, or in close proximity to, group companions.

Factors determining likelihood of being picked up by group companions. The social groups in which the infants lived differed in composition, and thus presumably in the extent to which group companions were predisposed to pick the infants up. Nevertheless two types of comparison proved interesting: data from first separations only were used.

First, the pre-separation values of certain measures were compared between infants which did and did not go on social companions during the separation period. Although all differences failed to reach significance, those infants which subsequently went on group companions tended, before separation, to be off their mothers less, to spend fewer ½ min more than 60 cm from their mothers, and be recorded sitting less often

when off ($P<0{\cdot}10$, Mann–Whitney U Test, two-tailed).

Second, for those infants which were recorded on group companions during the separation period, Spearman rank order correlation coefficients between the amount of time spent on group companions and various pre-separation measures were calculated. Those infants which spent most time on group companions tended to be those which before separation were responsible for the highest percentages of approaches and leavings, showed most locomotor activity (no. boxes per ½ min), were most active when active (i.e. $\frac{2\text{–}4}{>1}$ per cent least), and sat least ($P<0{\cdot}05$, one-tailed). There emerges, therefore, a general picture that those infants which were most active before separation tended to be on group companions most during separation.

One other correlation is worth mentioning. There was no difference between infants which did and did not go on group companions in the rate of whoo-calling before or during separation. However, amongst those which did go on group companions, those which went on most tended to be those which during separation whoo-called most when off ($P<0{\cdot}05$, one-tailed). One possible explanation of these results is that, in this situation, whoo-calling did not affect the probability that infants would go on group companions but increased the readiness of a generally accepting group companion to hold them.

The effect of being held by group companions on whooing rates. Whoo-calls are given when the mother is absent and when she rejects her infant. The infant also calls while on the mother, but much less frequently than when off her: this is shown in Table V. We may therefore ask whether the calling rate is similarly reduced when the infant is held by a group companion other than the mother.

Table VI shows the number of animals which called more when off all animals than when on group companions other than the mother. The number of animals which went on group companions is small, but the data show that being held by a group companion other than the mother does reduce the rate of whoo-calling, though perhaps not so consistently as does being held by the mother. Whether the mother is in fact more effective than other group companions in reducing the rate of whoo-calling can only be shown by a comparison between rates of calling on mother and on group companion in the same period. Here the scanty data available (Table VII) suggest little consistency in the relative effectiveness of mother and other group companions.

Table V. Number of Animals Whose Whoo-Calling Rate was More Frequent when off Mother than when on, as a Proportion of all Animals for whom Data is Available

		Pre-separation	Day 10	Days 11–16	Days 23–37
First separation	No.	11/16(5=0)	13/14(1=0)	15/16(1=0)	13/15(2=0)
	$P<$	0·002	0·002	0·002	0·002
Second separation	No.	8/10(2=0)	9/9	10/10	7/10(3=0)
	$P<$	0·008	0·004	0·002	0·016

Significance levels from Sign test, two-tailed.

Table VI. Number of Animals which Whoo-Called More when off all Animals than when on Group Companions, as a Proportion of Number of Animals which were on Group Companions

		Pre-sep.	Day 4	Separation mean	Day 10	Days 11–16	Days 23–37
First separation	No.	1/2	5/7	9/9	4/6	6/7	5/5
	$P<$	NS	NS	0·004	NS	NS	NS
Second separation	No.	2/5	1/1	4/4	4/4	3/4	1/2
	$P<$	NS	NS	NS	NS	NS	NS

Significance levels from Sign test, two-tailed.

Table VII. Number of Animals which Whoo-Called less when on Mother than when on Group Companions in the same Period, as a Proportion of Number which were on Group Companions

		Pre-separation	Day 10	Days 11–16	Days 23–37
First separation	no.	1/2	3/6(1=0)	2/7(1=0)	1/5(3=0)
Second separation	no.	1/2(1=0)	0/4(3=0)	1/4(3=0)	1/2(1=0)

Does being held by a group companion affect the consequences of the separation experience? If being held by a group companion reduces the rate of whoo-calling in the same way as does being held by the mother, it might also affect the longer-term consequences of maternal deprivation. The same methods of analysis as used in the previous sections have been applied here to data for the separation period, days 11 to 16 and days 23 to 37.

During the separation period, infants which were sometimes on group companions tended to be recorded sitting less than those which were not, but neither this nor the differences between the other activity measures reached significance. Among the nine infants which were on group companions, those which were on them most showed most locomotor activity and were most active when active ($P<0{\cdot}05$), and tended to sit least ($r_S = -0{\cdot}50$, non-significant). In these respects their behaviour resembled that in the pre-separation period.

On days 11 to 16 and 23 to 37 none of the differences between infants which had and had not been on group companions during the separation period reached significance, but those which had, tended to be more active ($P<0{\cdot}10$, Mann Whitney U test, two-tailed) in the latter period. Among those which had been on group companions, there was a tendency for those which spent most time at a distance from their mothers to be on group companions most: correlations between time on group companions and both the number of ½ min in which they went more than 60 cm and the number they spent more than 60 cm from their mothers were all positive, that for the latter reaching significance ($P<0{\cdot}01$) in the latter period. Apart from this there were, as previously, positive correlations with activity measures (%Ap., %L., no. boxes per ½ min), though only that with per cent Ap. in days 23 to 37 reached significance ($P<0{\cdot}01$). The animals which had been on group companions also tended to be most active when active ($P<0{\cdot}05$ in days 11 to 16).

In addition to possible effects of having been on group companions on the absolute values of measures during the post-separation period, we examined changes in certain of the measures between pre- and post-separation periods. In days 23 to 37 those animals which had not been on group companions showed a greater increase in %Ap.–%L than did infants which had ($P<0{\cdot}02$, two-tailed): this is the only suggestion that having been on group companions ameliorated the consequences of the separation period.

In conclusion, there were few differences during the post-separation period which could be related to being on group companions during separation and, apart from the change in the infant's role in maintaining proximity, those which did occur were similar in nature to those occurring before separation. They were thus more likely to be causal to interaction with group companions than a consequence of it.

Age at Separation

Infants were separated for the first time at 21, 25 to 26 or 30 to 32 weeks. Here the absolute scores, and the changes from pre-separation values, are compared for each pair of age-groups (Mann–Whitney U test, two-tailed).

21- vs 30- to 32-week groups. (a) *Proportion of time on or off mother.* Before separation the 21-week group spent less time awake on the nipple than the 30-week group; this difference cannot be accounted for by an age effect according to results obtained from control infants (Hinde & Spencer-Booth 1967). Thereafter the only significant difference was that on days 11 to 16 the 21-week infants were asleep more than the 30-week ones (a trend which was apparent throughout the experiment). There were no differences in the percentage change over pre-separation in any of these measures.

(b) *Responsibility for time off.* The only significant difference between absolute scores was that on day 10 the 21-week infants were rejected (R) less frequently than the 30-week infants. However on day 10 the percentage decrease in the frequency of rejections was greater for the

21-week than the 30-week infants ($P<0{\cdot}02$). The relative frequency of rejections (R/(A+M+R)) decreased in the 21-week group but increased in the 30-week ($P<0{\cdot}01$) both on day 10 and days 23 to 37.

Thus mothers were apparently more tolerant of the increased demands of 21-week infants than they were of 30-week infants. Since there was no significant difference in time off or in the decrease in time off between the two groups, it seems that the 30-week infants were attempting to get on their mothers more than the 21-week infants.

(c) *Distance when off.* No significant differences were found between groups.

(d) *Responsibility for proximity.* No significant difference in any of the absolute scores of these measures was found. Comparing percentage changes, the 21-week group showed a smaller decrease in the percentage of leavings for which the infant was responsible than did the 30-week group. This again suggests a greater role in maintaining proximity amongst the 30- to 32-week infants, and their persistently (but not significantly) higher scores on %Ap.–%L are in harmony with this.

(e) *Vocalizations.* Only noises made whilst off other animals are considered here. The median whoo-calling rate was significantly higher for 30- to 32-week infants than for the 21-week ones (differences significant for pre-separation and separation means, $P=0{\cdot}54$, and for days 23 to 37, $P=0{\cdot}032$). Whoo-calling before separation was too infrequent for comparisons of percentage changes to be useful.

Neither tantrums nor squeaks/geckers/screeches showed any differences in absolute scores or, where this could be assessed, in percentage changes.

(f) *Locomotor activity.* The 21-week infants were generally less active than the 30-week ones, but the only difference which nearly reached significance was in $\frac{2\text{–}4}{>1}$ per cent for the separation mean ($P=0{\cdot}056$). There were no differences in percentage change.

(g) *Type of activity.* There were no significant differences in either absolute scores or percentage change.

21- vs 25- to 26-week groups. (a) *Proportion of time on or off.* The 21-week infants tended to be awake on the nipple more than the 25- to 26-week infants (difference significant for pre-separation mean and for days 11 to 16, $P=0{\cdot}054$ and $0{\cdot}02$, respectively). The former group also tended to have a higher on and off score throughout the experiment, but this was only significant on days 23 to 37. There were no significant differences in percentage changes.

(b) *Responsibility for time off.* On day 10 the 21-week infants were rejected significantly less than the 25-week infants ($P=0{\cdot}052$), and the 21-week infants had a median decrease in days 11 to 16 while the 25-week infants had a median increase ($P=0{\cdot}02$).

(c) *Distance when off.* The only difference which reached significance was that the 21-week group leant against the mother less than the 25-week group ($P=0{\cdot}052$).

(d) *Responsibility for proximity.* No differences reached significance.

(e) *Vocalizations.* The 21-week infants tantrummed less than the 25- to 26-week infants in all periods (significant for days 23 to 37, $P=0{\cdot}02$). Control infants tended to tantrum more from weeks 20 to 25. No differences for other vocalizations reached significance.

(f) *Locomotor activity.* No significant differences were found.

(g) *Type of activity.* No significant differences were found.

25- to 26- vs 30- to 32-week groups. (a) *Proportion of time on or off.* Before separation the 25-week infants were awake on nipple less than the 30-week ones ($P=0{\cdot}008$) and on days 11 to 16 they were both awake on nipple and on mother off nipple less ($P=0{\cdot}02$ and $0{\cdot}056$, respectively). There were no differences between the groups in percentage change in any of these measures.

(b) *Responsibility for time off.* Before separation the 25-week infants received a higher proportion of rejections than the 30-week infants ($P=0{\cdot}032$). After separation, the absolute scores were no different, and it follows that the percentage change tended to decrease for the 25-week infants, whereas for the 30-week infants it increased ($P=0{\cdot}056$ on day 10; $0{\cdot}02$ on days 23 to 37).

(c) *Distance when off.* No significant differences were found.

(d) *Responsibility for proximity.* Before separation the 25-week infants had a higher following mother (per cent F) score than the 30-week infants ($P=0{\cdot}032$), and this difference was maintained after separation, but was significant only on day 10 ($P=0{\cdot}008$). Thus the 25-week infants were playing a greater role in maintaining proximity with their mothers than

the 30-week infants; this is supported by their generally higher %Ap.–%L scores. Data from control infants suggest a tendency for %Ap.–%L to decline slightly from about weeks 20 to 30, so the differences here may have been unrelated to the separation. There were no differences between groups in percentage change.

(e) *Vocalizations.* Whoos and squeak/gecker/screeches were less for the 25-week infants than the 30-week ones (except on the day the mother returned), but the reverse was true for tantrums. The only significant difference was for tantrums on days 23 to 37 ($P=0{\cdot}032$).

(f) *Locomotor activity.* No significant differences were found.

(g) *Type of activity.* The 25-week infants tended to sit and fiddle more than the 30-week infants (difference significant on day 4, $P=0{\cdot}032$). However they showed less social play (difference significant on days 11 to 16 and 23 to 37, $P=0{\cdot}056$ and 0·03, respectively).

Conclusion. Less than 1 : 20 of the differences between the age groups were significant, and it is clear that age at separation was not an important variable over the range studied.

Effect of Repeated Separations

Both the 21-week group ($N=5$, one animal having died) and the 25 to 26 week group ($N=5$) were separated again at 30 to 31 weeks. Comparisons were made between the first and second separations of each group, between the two groups given a second separation in week 30 (21(30) and 25(30) groups), and between each of these and those separated for the first time in weeks 30 to 32.

First vs second separations of the same individuals. Differences between first and second separations were assessed using the sign test. No significant differences were found for either the 21 vs 21(30) or the 25 vs 25(30) week comparisons.

Comparison of second separation groups. There were some indications that the group first separated at 21 weeks was more affected by the second separation than the group first separated at 25 to 26 weeks. Although few differences were significant, the former group spent more time asleep ($P=0{\cdot}02$) and ate less ($P=0{\cdot}06$) on day 10, and had shorter periods off on days 23 to 37 ($P=0{\cdot}06$). They showed a greater fall in time at a distance on day 10 ($P=0{\cdot}03$), and in time off ($P=0{\cdot}06$) and time on mother off nipple ($P=0{\cdot}03$) on days 11 to 16.

First vs second separations at 30 weeks. (a) (21) 30- vs 30-week groups. *Proportion of time on or off.* There were no significant differences for the absolute measures. However the time asleep, which tended to be less for the 30-week infants than the (21) 30-week group, increased less on day 10 for the 30-week infants ($P=0{\cdot}032$). Time on mother off nipple and on and off increased with respect to pre-separation on days 11 to 16 for the 30-week group but decreased for the (21) 30-week group ($P=0{\cdot}032$).

Responsibility for time off. The 30-week infants were generally rejected less than the 21(30)-week group, but this was only significant (for R/(A+M+R)) on days 11 to 16 ($P=0{\cdot}056$). On days 23 to 37 the 30-week group were being rejected on a higher proportion of attempts than before separation and the (21)30-week group on a lower proportion ($P=0{\cdot}056$).

Time at a distance. There were no significant differences.

Responsibility for proximity. Median scores for the infants' role in maintaining proximity and for following the mother were lower for the 30-week group throughout, but the difference reached significance only on days 11 to 16 for the former ($P=0{\cdot}03$). The percentage change reached significance only for per cent Ap. on days 11 to 16, when the decrease was less for the 30-week infants than for the (21)30-group.

Vocalizations. Whoos and squeak/gecker/screeches were higher for the 30-week group than the (21)30 infants: the differences were significant for whoos on day 4 and the separation mean ($P=0{\cdot}02$) and for days 23 to 37 ($P=0{\cdot}008$); they were also significant for squeak/gecker/screeches during separation ($P=0{\cdot}032$). However the percentage increase in whoos during and after separation was generally greater for the (21)30's than the 30's. During separation the squeak/gecker/screech noises increased with respect to pre-separation for the 30-week infants, but decreased for the (21)30's ($P=0{\cdot}056$).

Locomotor activity. There were no significant differences.

Type of activity. The only significant difference was on days 23 to 37 when the per cent E score showed a smaller increase for the 30's than for the (21)30's ($P=0{\cdot}056$).

(b) (25) 30 vs 30 weeks. *Proportion of time on or off.* Before separation the 30-week group was awake on nipple more than the (25)30 infants ($P=0{\cdot}02$). After separation the former tended to be off all animals less, although never

significantly so. They were awake on nipple and on and off more on days 11 to 16 ($P=0{\cdot}032$), and on mother off nipple more on days 23 to 37 ($P=0{\cdot}056$). The percentage increase in time asleep was greater for the 30 than the (25)30 infants on day 10 ($P=0{\cdot}008$).

Responsibility for time off. The 30-week group were always rejected less frequently and on a smaller proportion of occasions, but the differences reached significance only for R/(A+M+R) on days 11 to 16 ($P=0{\cdot}032$). There were no differences in percentage change.

Distance when off. There was a tendency for the 30-week infants to be at a distance from their mothers more than the (25)30 infants, but this was significant only before separation ($P=0{\cdot}056$). Percentage change never differed significantly.

Responsibility for proximity, Per cent L by the 30-week infants was always higher than that by the (25)30 infants; %Ap.–%L was lower but significantly only on days 11 to 16 ($P=0{\cdot}032$). The percentage change never differed significantly.

Vocalizations. Whoos were significantly lower for the (25)30 infants than for the 30-week infants on day 4 and during separation ($P=0{\cdot}008$). The same was true of squeak/geckers/screeches ($P=0{\cdot}032$ and $0{\cdot}056$, respectively). However the (25)30's tantrummed more than the 30's before separation ($P=0{\cdot}056$).

The 30-week infants increased their rate of S/G/S during separation, but the (25)30's decreased ($P=0{\cdot}032$ for day 4; $P=0{\cdot}008$ for separation mean).

Locomotor activity. There were no significant differences in absolute measures, although the (25)30's tended to be less active (lower no. boxes per $\frac{1}{2}$ min; higher $\frac{2\text{–}4}{>1}$ per cent).

The percentage increase in sitting was higher during separation for the 30's than the (25)30's.

Type of activity. There were no significant differences in absolute measures, and the only percentage change difference was that the 30's showed a decrease in eating (per cent E) on days 11 to 16 and the (25)30's an increase ($P=0{\cdot}056$).

Conclusion. In summary, there were few significant differences between the animals separated for the first time at 30 weeks and those separated for a second time at that age, though on some measures the latter group appeared to be less disturbed by the separation experience than the former.

The Effect of Separation on the Response to External Stimuli

Responses to strange objects and mildly frightening situations were tested as described in Spencer-Booth & Hinde (1967), during both first and second separation experimental periods. During separation the number of visits to and the total time in the 'filter' cage containing the model toy decreased with respect to pre-separation in a significant proportion ($P<0{\cdot}01$ and $0{\cdot}05$, respectively) of first separation infants, but not second. There were no differences between pre- and post-separation tests. The time spent eating decreased more (or increased less) in first separation infants in the post-separation than in the pre-separation tests, and time spent sitting showed the converse effect ($P<0{\cdot}05$ for both measures).

Discussion

These results confirm and extend the results of the four separation experiments described in our earlier paper (Spencer-Booth & Hinde 1967). In general, the effects of separation can be described as follows. The behaviour of the infant, as shown by its vocalizations and locomotor activity, was immediately affected by the mother's removal. Whoo vocalizations increased markedly in all infants. The changes in activity were more complex: immediately after the mother's removal some infants alternated periods of hyperactivity with periods of sitting inactive and thus showed little change in overall locomotor activity, while in others overall locomotor activity declined almost straight away. In either case, within a day or two locomotor activity, as well as specific activities such as play, became depressed. Although the whoo-calling induced by the mother's removal decreased through the 6 days of the separation period, the depression of locomotor activity tended to increase. While the mother was away the infant sought proximity with other animals in the pen, and in some cases was cuddled by them.

When the mother was returned, the infant sought contact with her vigorously, and spent more time on or near her than it had done before separation. In spite of a tendency, significant only in the younger infants, for the mother to reject the infant less than before separation, tantrums increased markedly. During the next few weeks whoo-calling and tantrums became less frequent and the times which the infant spent on or near its mother returned towards their pre-separation levels.

The disappearance of the effects of separation occurred at markedly different rates in different infants, and in different measures in the same infant. Furthermore the recovery appeared to be far from smooth, indicating that the readjustments of the mother and infant to one another after reunion were complex. For instance many mothers changed from being initially more permissive towards their infants than before separation to being temporarily or permanently more rejecting. Such changes in the mother may be induced at least partly by the 'excessive' demands of the infant, and our observations do not exclude the possibility that peripheral factors, such as sore nipples, played a role. The complexity of the changes is discussed further by Hinde & Spencer-Booth (1971): it seems likely that the nature of mother–infant interaction after reunion would affect the subsequent course and extent of the infant's recovery.

An outstanding characteristic of these data is the marked individual variability in the response to separation. While the directions of the effects of most of our measures were consistent, their extent was not predictable on any obvious grounds. Thus most differences between the sexes did not reach significance, though a number of measures indicated that males were somewhat more upset than females. Contact with group companions during the separation period, itself only tenuously related to pre-separation or separation behaviour, influenced the amount of whoo-calling during separation; but most of the post-separation differences between infants which had and had not had contact with group companions during separation appeared to be a reflection of pre-separation individual differences. Within the range studied here, age at separation also appeared not to be an important variable, except that the 21-week infants were rejected relatively less on reunion than were the older infants. In view of these facts the individuality of the response must stem from more subtle aspects of the infants' characteristics and of its relationship with its mother before separation. A further analysis of these data, concentrating on individual differences in the mother–infant relationship, is presented elsewhere (Hinde & Spencer-Booth 1971).

Infants separated for the second time showed the same kinds of responses as infants separated for the first time at the same age, and there was little evidence that the experience had either a more or a less severe effect the second time. However, although the once-separated infants seemed no different from the twice-separated infants in the first few weeks after separation, this does not necessarily mean that there were no longer-term effects of repeated separation which may be revealed by later observations on these infants.

Bowlby (1961), discussing the behaviour of human children who had lost their mothers, assigned the responses to three broad phases: 'protest', 'despair' and 'detachment'. The first of these, in which the children cry a great deal, is followed by a phase in which their behaviour becomes disorganized, and they are restless, apathetic and depressed. In the third phase Bowlby describes the behaviour as becoming reorganized on the basis of the mother's permanent loss. Bowlby was mainly concerned with children who had been separated from their mothers by going into hospital, and thus also entered a strange environment. Although the rhesus infants described here were not in a strange environment during separation their responses were similar in several respects. Thus the immediate response was one of greatly increased whoo-calling, accompanied by much more frequent sitting doing nothing than before separation. As these two responses became less extreme over the following few days, the infants appeared to become very depressed: they were progressively less active and less playful than before separation. Like the separated pigtail (*Macaca nemestrina*) infants described by Kaufman & Rosenblum (1967b), they thus seemed to show the symptoms of Bowlby's first two phases, although with considerable overlap between them. There was little if any evidence of any reorganization of behaviour by the rhesus infants, although possibly the gradual reduction in the increased rate of sitting doing nothing, and the increase in the time spent eating during the separation period, could be thought of as a reorganization in the animal's pattern of activity. Again, however, these changes overlapped the others. Possibly reorganization might become more apparent with a longer separation period, since Kaufman & Rosenblum (1967b) found that the intense depression in pigtail infants only began to lift after 5 to 6 days. Be that as it may, within the 6 days of separation there were progressive and interwoven changes in rhesus infants subjected to maternal loss, at least qualitatively similar to some of those seen in human children. The intensely 'clinging' behaviour shown by the rhesus infants in the immediately post-separation period is also

characteristic of many human children, though some of the latter after severe separations show instead 'detachment' and a difficulty in re-establishing affectional relations which we did not see in these monkeys.

Acknowledgments

This work was supported by the Medical Research Council and the Royal Society. We are grateful to R. Pollard-Urquhart, S. Atkinson, S. Richards and L. Davies for assistance in collection of the data, and to members of the Department of Veterinary Clinical Studies, Cambridge, especially A. R. Jennings and D. E. Bostock, for advice and help.

REFERENCES

Ainsworth, M. D. (1962). The effects of maternal deprivation: a review of findings and controversy in the context of research strategy. In: *Deprivation of Maternal Care.* Geneva: World Health Organization.

Bowlby, J. (1952). *Maternal Care and Mental Health,* 2nd edn, Monograph Series, No. 2. Geneva: World Health Organization.

Bowlby, J. (1961). Processes of mourning. *Int. J. Psychoanal.,* **42,** 317–340.

Hinde, R. A. & Atkinson, S. (1970). Assessing the roles of social partners in maintaining mutual proximity, as exemplified by mother–infant relations in rhesus monkeys. *Anim. Behav.,* **18,** 169–176.

Hinde, R. A. & Rowell, T. E. (1962). Communication by postures and facial expressions in the rhesus monkey (*Macaca mulatta*). *Proc. zool. Soc. Lond.,* **138,** 1–21.

Hinde, R. A. & Spencer-Booth, Y. (1967). The behaviour of socially living rhesus monkeys in their first two and a half years. *Anim. Behav.,* **15,** 169–196.

Hinde, R. A. & Spencer-Booth, Y. (1971). Individual differences in the responses of rhesus monkeys to a period of separation from their mothers. *J. Child Psychol. Psychiat.,* (In press).

Hinde, R. A., Rowell, T. E. & Spencer-Booth, Y. (1964). Behaviour of socially living rhesus monkeys in their first six months. *Proc. zool. Soc. Lond.,* **143,** 609–649.

Jensen, G. D. & Tolman, C. W. (1962). Mother–infant relationship in the monkey, *Macaca nemestrina*: the effect of brief separation and mother–infant specificity. *J. comp. physiol. Psychol.,* **55,** 131–136.

Kaufman, I. C. & Rosenblum, L. A., (1967a). Depression in infant monkeys separated from their mothers. *Science, N.Y.,* **155,** 1030–1031.

Kaufman, I. C. & Rosenblum, L. A. (1967b). The reaction to separation in infant monkeys: anaclitic depression and conservation–withdrawal. *Psychosomat. Med.,* **29,** 648–675.

Kaufman, I. C. & Rosenblum, L. A. (1969). Effects of separation from mother on the emotional behaviour of infant monkeys. *Ann. N.Y. Acad. Sci.,* **159,** 681–695.

Lebovici, S. (1962). The concept of maternal deprivation: a review of research. In: *Deprivation of Maternal Care.* Geneva: World Health Organization.

Rosenblum, L. A. & Kaufman, I. C. (1968). Variations in infant development and response to maternal loss in monkeys. *Am. J. Orthopsych.,* **38,** 418–426.

Seay, B. & Harlow, H. F. (1965). Maternal separation in the rhesus monkey. *J. nerv. ment. Dis.,* **140,** 434–441.

Seay, B., Hansen, E. & Harlow, H. F. (1962). Mother–infant separation in monkeys. *J. Child Psychol. Psychiat.,* **3,** 123–132.

Spencer-Booth, Y. (1968). The behaviour of group companions towards rhesus monkey infants. *Anim. Behav.,* **16,** 541–557.

Spencer-Booth, Y. & Hinde, R. A. (1967). The effects of separating rhesus monkey infants from their mothers for six days. *J. Child Psychol. Psychiat.,* **7,** 179–197.

(*Received* 26 *June* 1970; *revised* 6 *September* 1970; *MS. number:* 990)